Experiment ‘p_4_3_d_count_vowels’ Results

June 4, 2026

Experiment outcome: SUCCESS
Bad responses: 0
Responses containing `assume` **:** 0
Responses containing `expect` **:** 0
Resolution attempts: 1
Hard fails (resolution): 0
Soft fails (resolution): 0
Verification attempts: 1

Problem Specification

Problem name: p_4_3_d_count_vowels
Natural language statement: Write a method that takes a string and returns the number of vowels in the string.
Method signature: p_4_3_d_count_vowels(s: string) returns (count: int)

Ensures

- `count == count_vowels(s)`

Functional Code Given

```
function count_vowels(s: string): int {
  if s == "" then 0
  else if s[0] in ['a','e','i','o','u','A','E','I','O','U'] then count_vowels(s[1..]) + 1
  else count_vowels(s[1..])
}
```

GenAI interactions

Below you will find all interactions between the 'user' (program) and the 'assistant' (OpenAI).

Program → GenAI

```
You are given the following task to perform in Dafny:

Write a method that takes a string and returns the number of vowels in the string.

The signature should be:

p_4_3_d_count_vowels(s: string) returns (count: int)

The method should respect the following contract:

ensures count == count_vowels(s)

The contract uses the following dafny code:

function count_vowels(s: string): int {
  if s == "" then 0
  else if s[0] in ['a','e','i','o','u','A','E','I','O','U'] then count_vowels(s[1..]) + 1
  else count_vowels(s[1..])
}

These function(s) must not be used in your implementation of the method.

Produce and show only the Dafny body of this method, including the curly braces that
    ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
    ↪  'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-Dn5TH0dsw5lnRUTpRXPss3MMVoCPr
Model: gpt-5.5-2026-04-23
Created at: 1780591475
Finish reason: stop
Usage: 460 tokens in, and 631 tokens out

```
{
  if s == "" {
    count := 0;
  } else {
    var restCount := p_4_3_d_count_vowels(s[1..]);
    if s[0] in ['a','e','i','o','u','A','E','I','O','U'] {
      count := restCount + 1;
    } else {
      count := restCount;
    }
  }
```

```
}
```

Final Program

```
function count_vowels(s: string): int {
  if s == "" then 0
  else if s[0] in ['a','e','i','o','u','A','E','I','O','U'] then count_vowels(s[1..]) + 1
  else count_vowels(s[1..])
}

method p_4_3_d_count_vowels(s: string) returns (count: int)
       ensures count == count_vowels(s)
{
  if s == "" {
    count := 0;
  } else {
    var restCount := p_4_3_d_count_vowels(s[1..]);
    if s[0] in ['a','e','i','o','u','A','E','I','O','U'] {
      count := restCount + 1;
    } else {
      count := restCount;
    }
  }
}
```

Total Token Usage

Input tokens: 460
Output tokens: 631
Reasoning tokens: 512
Sum of 'total tokens': 1091

Experiment Timings

Overall Experiment started at 1780591469557, ended at 1780591545557, lasting 76000ms (76.00 seconds)
Iteration #1 started at 1780591469566, ended at 1780591545556, lasting 75990ms (75.99 seconds)